ANALOG CIRCUITS AND SENSOR SYSTEMS

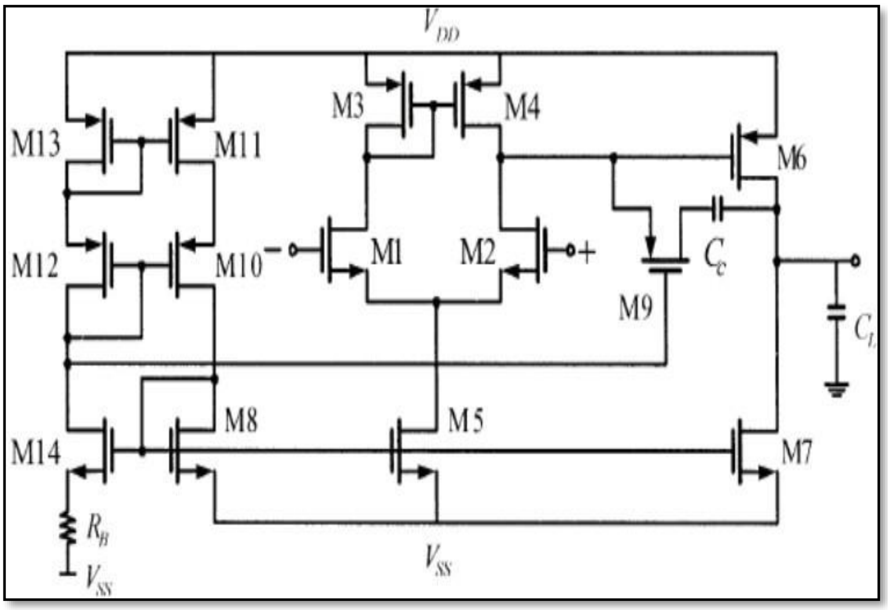
PROJECT # 01

DESIGN OF TWO STAGE OPAMP

WAJEEHA KAZMI

MATRICOLA # 0001177018

SCHEMATIC: 2-STAGE OPAMP



SPECIFICATIONS:

VDD/VSS	2.5/-2.5 V
Slew Rate	+5/-5 V/μs
Load Capacitor C_L	5pF
Max Vin Common Mode	2.1V
Min Vin Common Mode	-1.3V
Max Vout	2.2V
Min Vout	-2.2V
Gain Bandwidth	5MHz
Gain Bandwidth (ω)	31.415 Mrad/sec
Differential Gain	>80dB
Phase Margin	>60º
Noise Voltage	38 nV/$\sqrt{Hz}$
KT	$4.11e^{-21}$
Vtn/Vtp	0.71/-0.901
β'_n/ β'_p	$182e^{-6}/41.5e^{-6}$
μ_n/ μ_p	0.0506/0.0115

TRANSISTOR SIZES:

S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**
0.666	**0.666**	**3.289**	**3.289**	**4.583**	**33.351**	**23.235**
S8	**S9**	**S10**	**S11**	**S12**	**S13**	**S14**
4.583	**6.578**	**6.578**	**6.578**	**6.578**	**6.578**	**4.583**

DESIGN PROCEDURE:

1- The first and foremost step in the design procedure is the determination of Capacitance Cc:

$$C_c = \frac{16\,KT}{3\,GB(\boldsymbol{\omega})\,Sn(f)}[1 + \frac{SR}{GB(\boldsymbol{\omega})\,V_{ov_3}}]$$

To calculate that, we need to first find the V_{ov3}:

$$V_{ov_3} = V_{DD} - V_{INCM}^{MAX} + Vt_n - |Vt_p|$$

2- Now, we can start sizing the transistor from M_6:

$$I_6 = I_7 = SR(C_c + C_L)$$

3- V_{ov6} can be calculated by this formula;

$$V_{ov_6} = V_{DD} - V_0^{MAX}$$

$$L_6 = \sqrt{\frac{3}{2}\frac{\mu_p\,V_{ov_6}\,C_c}{GB(\boldsymbol{\omega})\,(C_c + C_L)\,tan(PM)}}$$

4- Now, we calculate W_6 and S_6 is equal to W/L;

$$W_6 = \frac{2\,I_6\,L_6}{\beta_p^{`}\,V_{ov_6}^2}$$

5- Next is to calculate I_5, and currents I_1=I_2=I_3=I_4:

$$I_5 = C_c\,SR$$

$$I_1 = I_2 = I_3 = I_4 = {}^{I_5}/_2$$

6- S_1 and S_2 are calculated;

$$S_2 = S_1 = \frac{2\,I_1}{\beta_n^{`}\,V_{ov_1}^2}$$

Where,

$$V_{ov_1} = \frac{SR}{GB(\boldsymbol{\omega})}$$

7- Calculate S_5 ($S_{5=}$ $S_{8=}$ S_{14}

$$S_5 = \frac{2\, I_5}{\beta_n^{`}\, V_{ov_5}^2}$$

$$V_{ov_5} = V_{i_{CM}}^{MIN} - Vt_n - V_{ov_1} - V_{SS}$$

8- Calculate S_7;

$$S_7 = \frac{C_c + C_L}{C_c}\, S_5$$

9- For perfect balancing, we need V_{SG6}= V_{SG4}=$V_{SG3.}$ Therefore, $S_{3,4}$ are calculated:

$$S_{3,4} = \frac{S_6}{2\, S_7}\, S_5$$

10- Calculate $S_{9:}$

$$S_9 = \frac{C_c}{C_c + C_L}\, S_6$$

11- For biasing network, we have

$$S_{10} = S_{11} = S_{12} = S_{13}$$

$$S_{12} = S_6 \, for \, I_{12} = I_6$$

$$R_B = 1/g_{m_8} = \frac{V_{ov_8}}{2\, I_8}$$

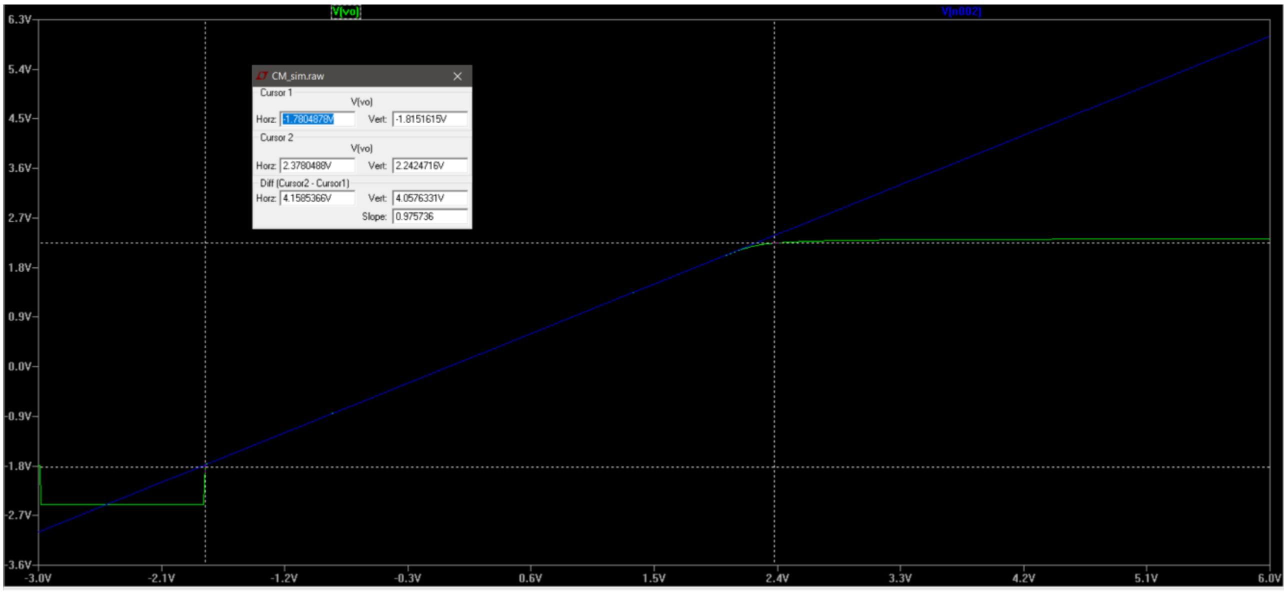
$$V_{ov_8} = \sqrt{\frac{2\, I_8}{\beta_n^{`}\, S_8}}$$

$$I_8 = I_7$$

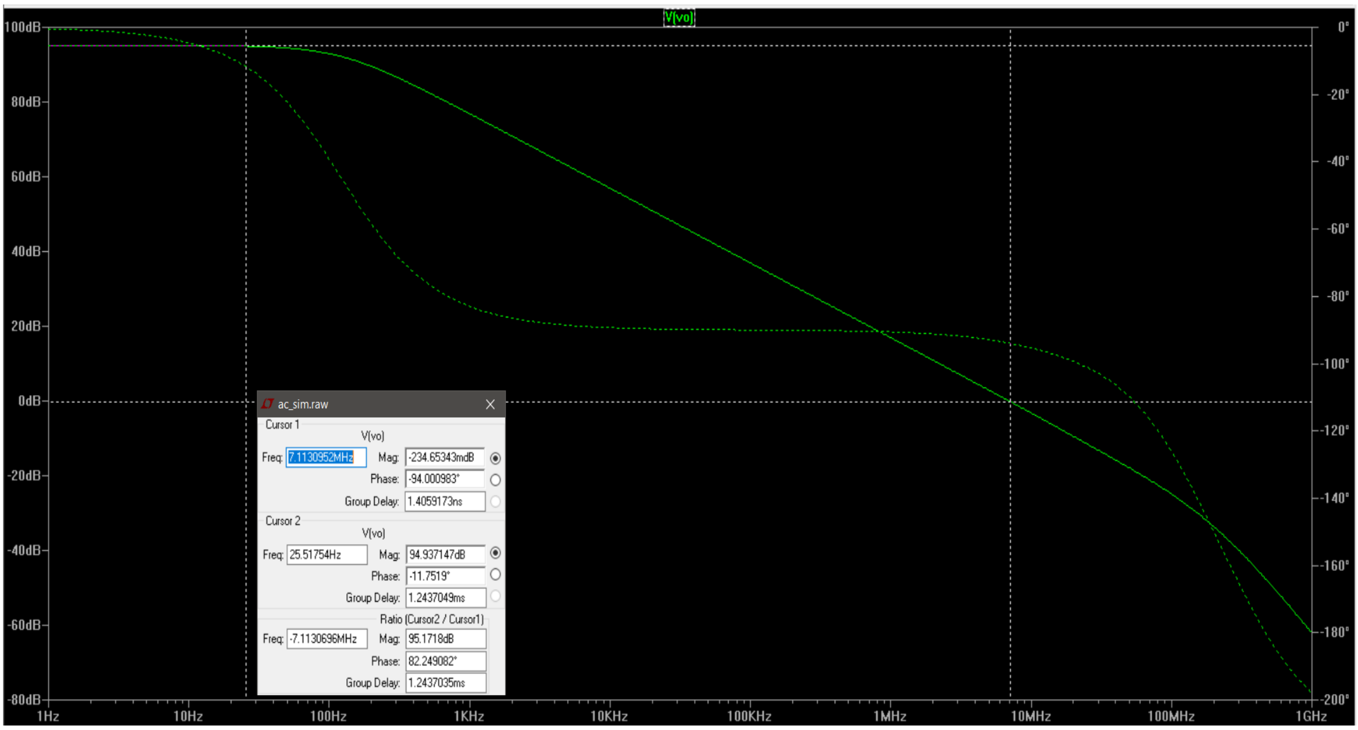
We proceed to implement it on LTSpice after determining the transistors' sizes. Before moving on to the complete circuitry, it is done to ensure that every transistor in the main circuit is in saturation under the current design conditions. After then we move on to the entire circuit after confirming that the transistors are in saturation. After creating the full circuit model including the biasing network and replacing the compensation resistance Rc with the transistor M9, we first again perform the .op simulation to verify that all of the transistors except M9 are in saturation.

RESULTS:

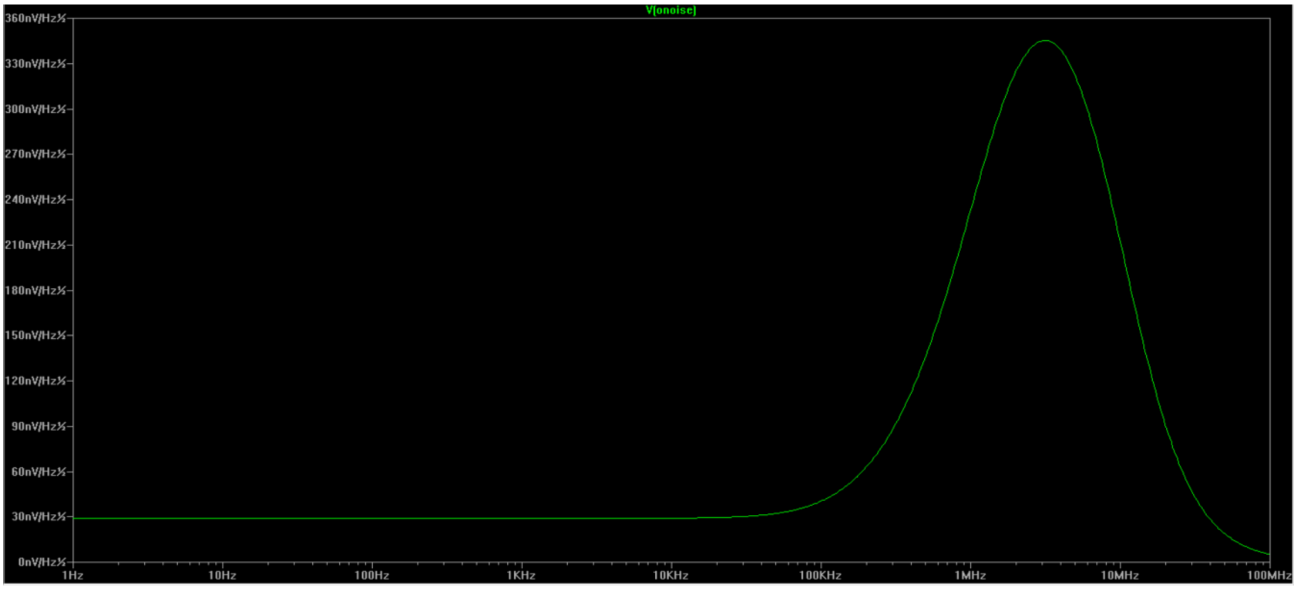
CMRR



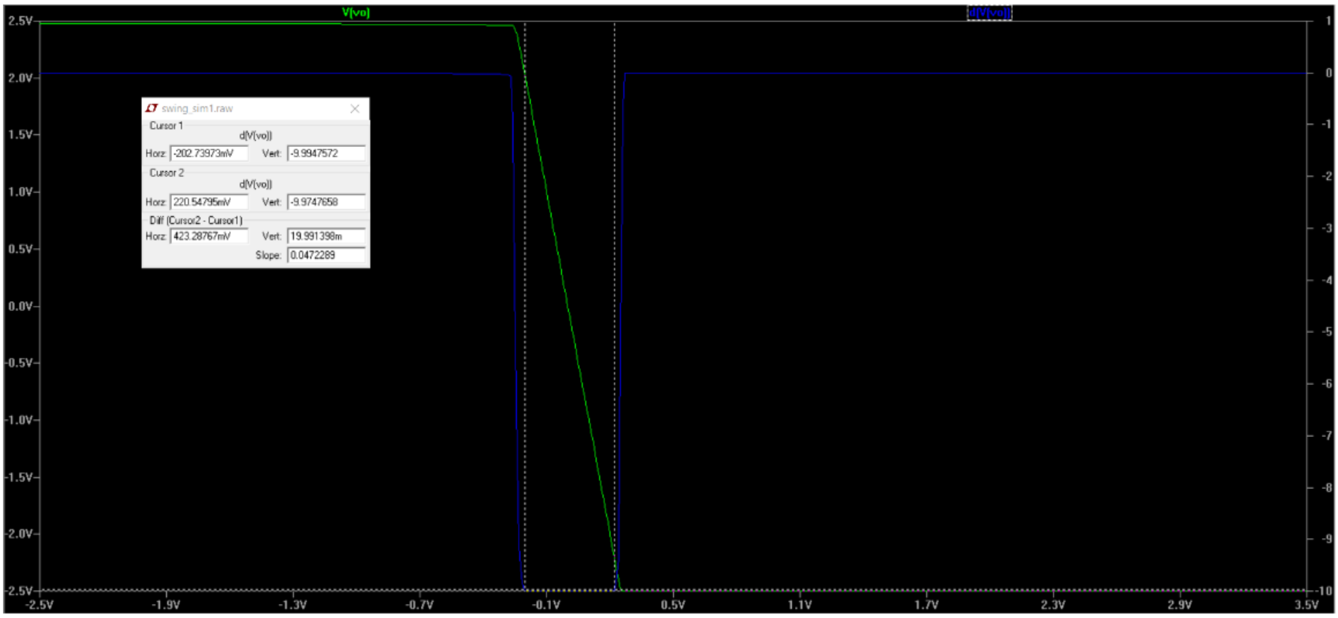
GAIN BANDWIDTH



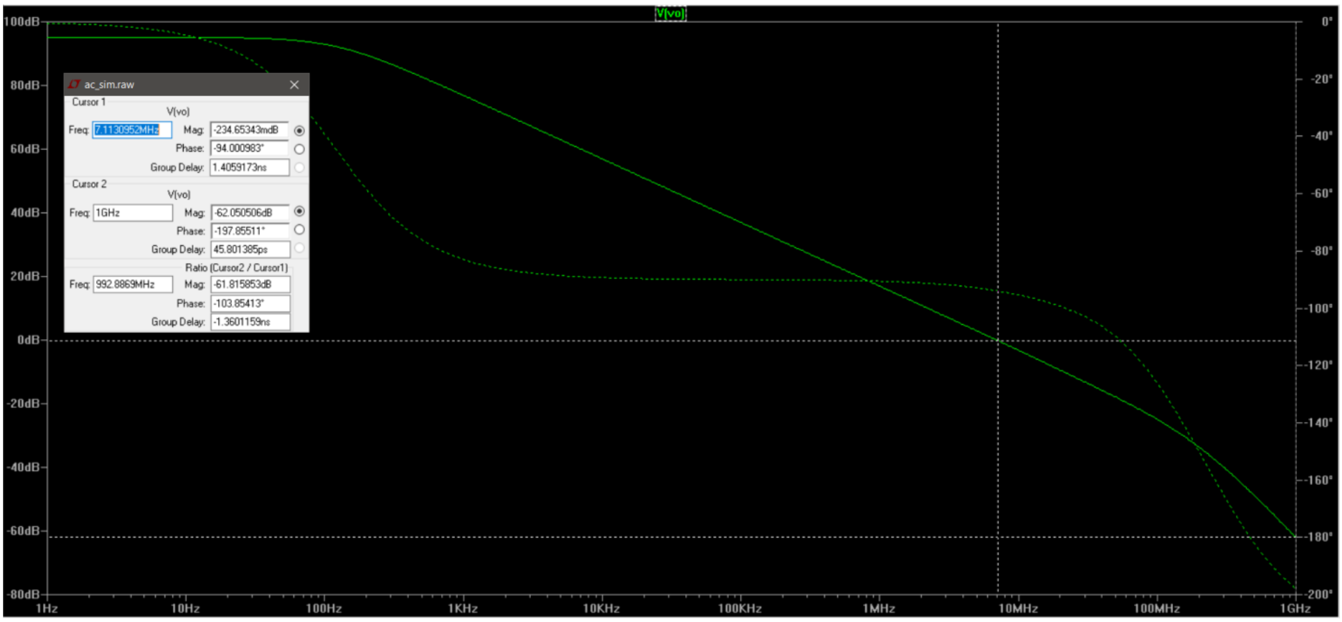
NOISE



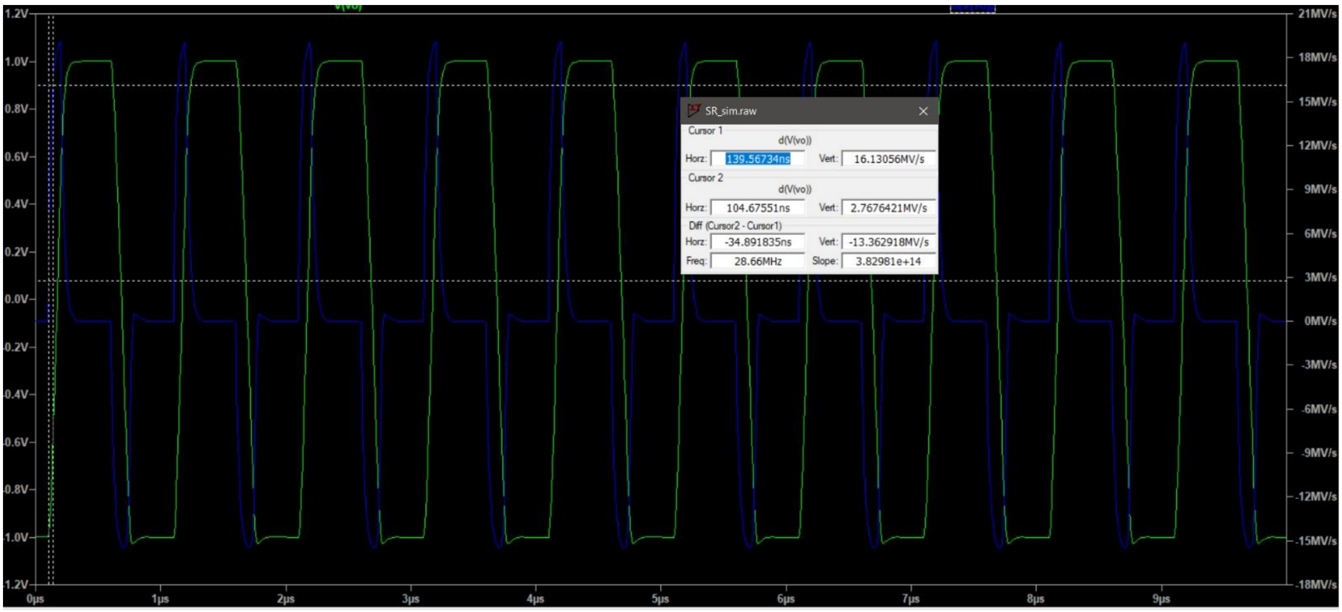
OPSWING



PHASE MARGIN



SLEW RATE



COMAPRISON:

PARAMETERS	GIVEN VALUE	ACHIEVED VALUE
VDD/VSS	2.5/-2.5 V	2.5/-2.5 V
Slew Rate	+5/-5 V/µs	+13/-13 V/µs
$V_{inCMmax}$	2.1V	2.2V
$V_{inCMmin}$	-1.3V	-1.8V
Max Vout	2.2V	2.5V
Min Vout	-2.2V	-2.5V
Gain Bandwidth	5MHz	7MHz
Differential Gain	>80dB	94dB
Phase Margin	>60º	83º